

MARKNESSE, Netherlands

WMO: 062730

Lat: 52.7019N

Lon: 5.8875E

Elev: -3

StdP: 101.36

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.6	-4.5	-9.6	1.6	-4.9	-7.1	2.1	-4.2	13.3	8.0	12.0	8.2	3.7	80	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.7	28.0	20.0	25.8	19.0	23.9	18.1	21.1	26.4	20.0	24.3	19.0	22.8	3.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.3	14.1	23.4	18.4	13.2	22.0	17.5	12.5	21.0	61.1	26.3	57.4	24.4	54.0	22.8	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
10.8	9.5	8.4	-9.0	32.5	4.9	1.8	-12.6	33.7	-15.4	34.8	-18.2	35.8	-21.7	37.1	
			DB												
			WB	-9.4	23.8	4.8	1.1	-12.9	24.6	-15.7	25.2	-18.4	25.9	-21.9	26.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	3.0	3.6	6.1	9.7	13.3	15.7	17.6	17.4	14.8	10.8	6.9	3.9
	DBStd	6.20	3.98	3.69	3.31	3.52	3.39	2.81	2.86	2.46	2.60	3.23	3.23	4.10
	HDD10.0	898	218	178	126	47	7	0	0	0	1	30	101	190
	HDD18.3	3008	477	411	381	258	161	89	49	48	111	234	343	447
	CDD10.0	996	1	1	4	39	109	171	236	228	144	54	7	2
	CDD18.3	63	0	0	0	0	4	10	27	18	4	0	0	0
	CDH23.3	526	0	0	0	9	53	93	216	127	27	1	0	0
	CDH26.7	123	0	0	0	1	7	22	59	30	4	0	0	0
Wind	WSAvg	4.2	5.1	4.8	4.7	4.2	4.2	3.7	3.4	3.3	3.5	4.1	4.3	4.9
Precipitation	PrecAvg	870	73	61	54	46	63	79	95	91	79	76	77	76
	PrecMax	1175	140	137	106	99	128	172	182	200	189	239	148	151
	PrecMin	655	8	20	11	0	29	33	40	11	9	18	11	25
	PrecStd	119	35	31	30	24	25	36	41	49	42	41	33	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	12.9	17.3	24.5	27.6	29.5	31.5	29.9	26.5	21.1	15.7	12.9
		MCWB	10.9	9.7	12.1	15.3	18.2	21.2	20.9	20.5	19.5	17.2	13.7	11.7
	2%	DB	10.5	10.9	14.7	20.9	24.6	26.0	28.1	26.7	23.5	18.1	13.9	11.7
		MCWB	9.5	9.2	10.7	13.6	16.0	19.2	20.3	20.1	18.6	15.5	12.6	10.6
	5%	DB	9.2	9.7	12.2	18.2	22.2	23.3	25.7	24.2	21.2	16.5	12.6	10.6
		MCWB	8.2	8.3	9.8	12.7	15.6	18.1	19.4	18.8	17.2	14.5	11.4	9.6
	10%	DB	8.2	8.5	10.7	15.8	19.7	21.2	23.4	22.3	19.3	15.3	11.4	9.1
		MCWB	7.2	7.3	8.8	11.6	14.6	16.9	18.6	17.8	16.2	13.6	10.3	8.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.0	10.6	12.9	16.2	19.7	22.2	22.6	22.4	20.5	17.5	14.1	12.0
		MCDB	11.9	12.1	15.7	22.5	25.0	27.7	28.2	27.7	25.1	20.3	15.1	12.7
	2%	WB	9.6	9.4	11.2	14.5	17.7	20.2	21.2	20.6	18.9	15.9	12.7	10.8
		MCDB	10.4	10.6	13.4	19.5	22.4	24.5	26.7	25.0	22.7	17.8	13.8	11.6
	5%	WB	8.4	8.4	10.1	13.2	16.2	18.8	20.2	19.4	17.7	14.9	11.5	9.6
		MCDB	9.1	9.5	12.1	17.3	20.4	22.5	24.7	23.2	20.5	16.3	12.5	10.4
	10%	WB	7.3	7.4	9.0	12.0	15.0	17.5	19.0	18.4	16.8	13.9	10.4	8.4
		MCDB	8.1	8.5	10.6	15.3	19.0	20.5	22.7	21.5	18.9	15.1	11.2	9.1
Mean Daily Temperature Range	5% DB	MDBR	3.9	4.9	6.6	8.8	8.8	8.6	8.7	8.5	8.0	6.6	4.8	3.9
		MCDBR	4.3	6.3	9.9	13.1	13.2	12.3	13.0	11.9	10.9	7.6	5.3	4.3
		MCWBR	4.0	5.0	6.7	7.1	6.3	6.9	6.3	6.2	6.3	5.2	4.4	4.0
	5% WB	MCDBR	4.3	5.7	8.8	11.4	10.9	11.2	11.6	10.6	9.6	7.3	5.2	4.3
		MCWBR	4.2	4.8	6.3	6.5	6.0	6.7	6.2	6.0	5.9	5.4	4.6	4.1
Clear-Sky Solar Irradiance	taub	0.332	0.352	0.381	0.407	0.409	0.406	0.418	0.411	0.389	0.366	0.344	0.325	
	taud	2.317	2.320	2.254	2.233	2.270	2.294	2.271	2.310	2.361	2.386	2.343	2.311	
	Ebn at Noon	642	746	797	824	842	847	828	812	784	721	619	575	
	Edh at Noon	63	83	108	124	126	124	125	114	97	77	61	53	
All-Sky Solar Radiation	RadAvg	0.65	1.33	2.55	4.16	5.10	5.38	5.15	4.29	3.06	1.74	0.77	0.50	
	RadStd	0.07	0.23	0.29	0.41	0.36	0.36	0.56	0.34	0.29	0.16	0.07	0.05	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (66 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page